

01 | Day 3: Scale It

How do the fundamentals become the systems used in modern AI?

Retrieve: Open your ACT-D2-04 evidence board and CHECK-D2-04 response. Do not edit them yet.

02 | Sort the Day 2 Evidence

Category	Example evidence
implementation	reset, mode, detached/frozen path, output/loss contract
optimization	update size, gradient flow, activation saturation
representation	effectively linear path, missing useful hierarchy
data/evaluation	weak signal, mismatch, leakage, invalid validation

Commit: Where does your leading Day 2 explanation belong? What would disconfirm it?

03 | The Day 3 Investigation

**depth -> valid data -> CNN mechanics -> training evidence -> rescue -> transfer -
> attention -> systems scale**

At every step:

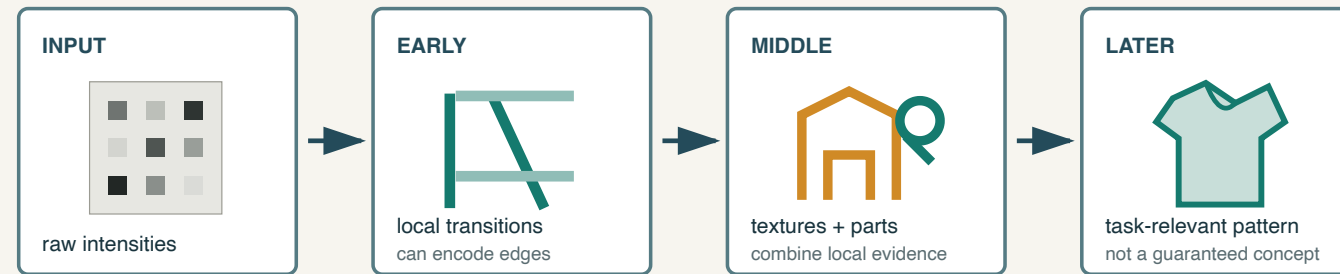
predict -> observe -> diagnose -> explain why -> state the limit

CORE: valid image workflow and evidence

04 | Depth Can Reuse Features

Depth can build reusable features, but it raises more than one curve

Read left to right. Layer roles are possibilities to test, not universal semantic labels.



Prediction prompt

Which evidence would show that added depth helped representation rather than only training fit?

Do not infer: deeper layers always become more abstract, human-readable, or more accurate.

05 | ACT-D3-01: Feature Ladder Under a Budget

Place each card at input, early, middle, or later:

- raw intensity;
- local transitions;
- textures/corners;
- sleeve/sole/collar arrangements.

Then choose: **add blocks | widen | audit the pipeline first**

Commit one expected observation and one rejection result.

06 | A Validation Score Has a Chain of Custody

For every feature or statistic, ask:

1. When was it available?
2. Which records created it?
3. Which split owned the fitted state?
4. Which decision used the result?

High score = clue, not proof of validity or leakage.

07 | ACT-D3-02: Leak Accusation Before Code

Evidence: validation AUC `0.997` ; a simple model nearly matches a larger one.

Submit before source access:

- two facts;
- two plausible explanations;
- one discriminating check;
- confirming and weakening evidence;
- first feature to quarantine.

A near-perfect score is not a unique fingerprint.

08 | LAB-D3-01 Launch

Find the Data Leak | 45 min

Predict: why the score is implausible.

Inspect: feature availability and fitted-state ownership.

Repair: split, training-only preprocessing, deployment-valid features.

Compare: leaky versus valid evidence under the same simple model.

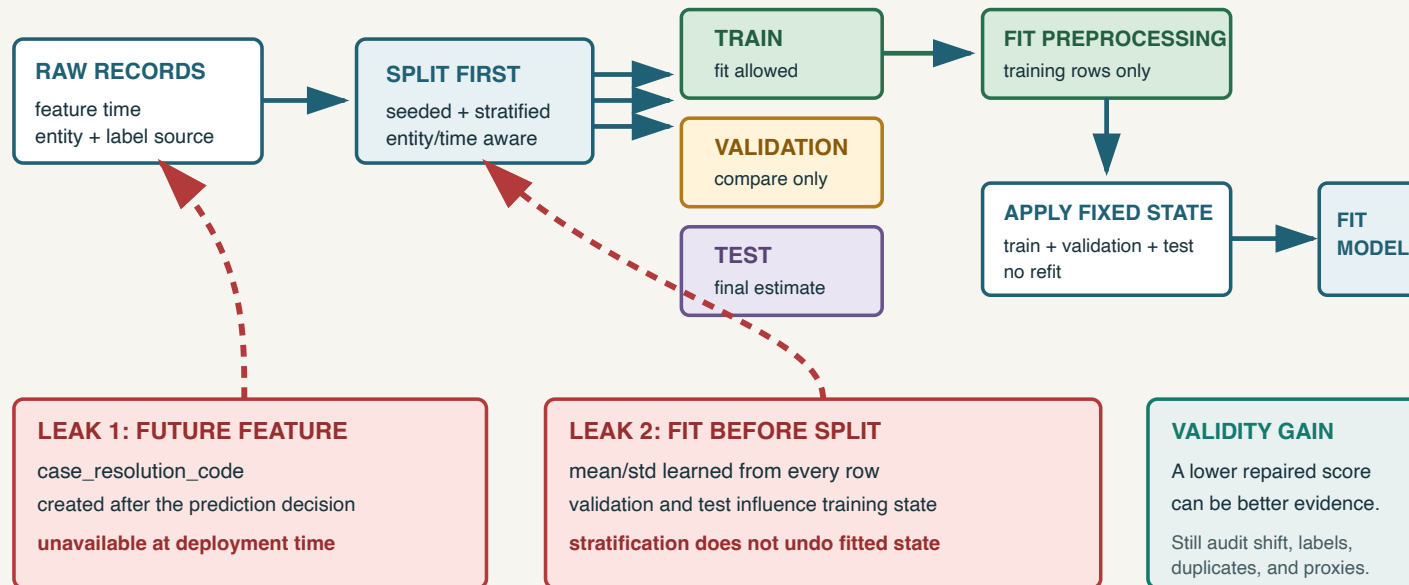
```
courseware/day-3/labs/LAB-D3-01-find-data-leak.ipynb
```

09 | LAB-D3-01 Debrief: Lower Can Be Better

POST-GATE REVEAL

Validation needs a chain of custody

Blue: valid ownership. Red: validation or future information crosses into training.



Prompt: Which information crossed a boundary, and when did it become available?

10 | Why Images Reward Locality

Flatten first:

- every location can receive independent weights;
- neighborhood structure is no longer explicit.

Convolve first:

- small local interactions;
- the same detector is reused across positions;
- channels keep spatial response maps.

Question: What does sharing buy, and what does it assume?

11 | ACT-D3-03: Predict the Kernel Map

Input has 0 values on the left and 4 values on the right.

```
-1  0  1  
-1  0  1  
-1  0  1
```

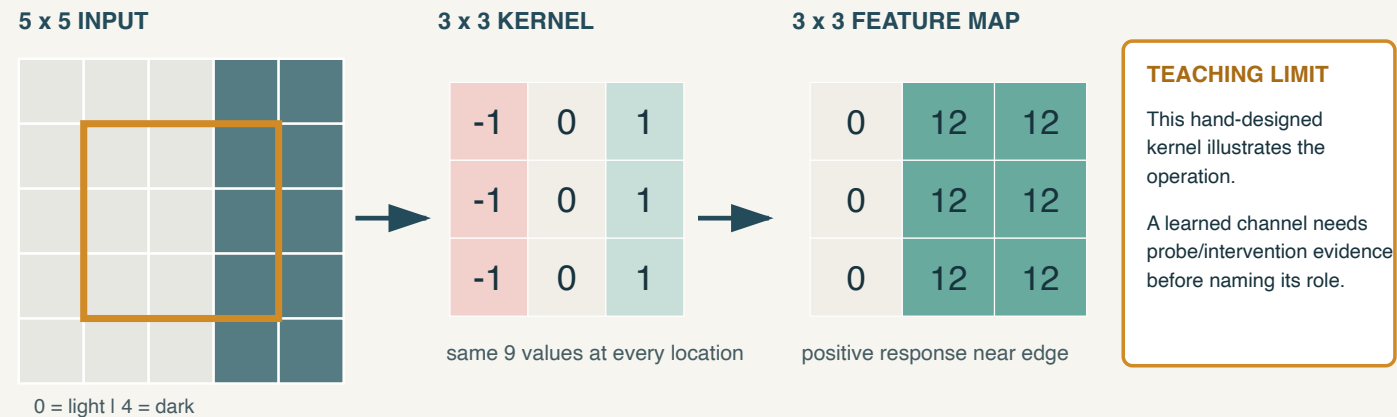
Stride 1, padding 0.

Commit: output shape, top-left value, largest-response location, and sign pattern.

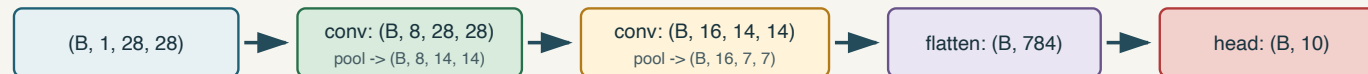
12 | One Patch, One Map, One Shape Trace

One shared local detector produces one feature map

Operation shown is cross-correlation: the kernel is not flipped.



Batch-first shape trace



Prediction prompt: Which dimensions change under pooling, and which stored weights are reused?

13 | LAB-D3-02 Launch

Compact CNN and Feature Maps | 60 min

Preflight: cache first; download only by explicit opt-in.

Predict: every `(B, C, H, W)` shape.

Repair: flattened dimension or channel order from printed evidence.

Observe: curves, selected maps, and misclassified examples.

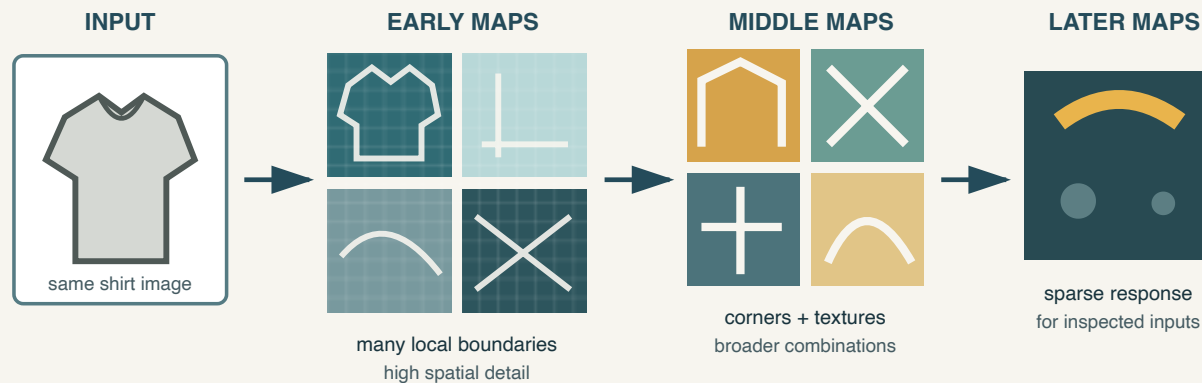
Limit: map brightness is descriptive, not causal proof.

```
courseware/day-3/labs/LAB-D3-02-cnn-feature-maps.ipynb
```

14 | Early Detail, Later Selectivity

Feature maps make internal responses visible, not self-explanatory

Compare the same input across stages. Describe first; interpret second; intervene before causal language.



SUPPORTED DESCRIPTION

"For these images, this later channel responds near the upper outline more strongly than elsewhere."

Name layer, channel, input set, and observed pattern.

NOT YET ESTABLISHED

"This is the shirt detector that caused the prediction."

Test with masking, translation, counterfactual inputs, and downstream-path evidence before causal or semantic claims.

Observation prompt: Which spatial detail disappeared, and what new selectivity is only a hypothesis?

15 | LAB-D3-02 Debrief: Describe Before Naming

Bring one example of each:

- a shape invariant that repaired the model;
- an early map observation;
- a later map observation;
- a confident mistake;
- an interpretation you would test with an intervention.

What did sharing buy? What remains unexplained?

16 | CHECK-D3-01 | 10 min

Trace:

```
(B,1,28,28) -> conv -> pool -> conv -> pool -> flatten
```

Then explain:

- parameter sharing;
- what batch size changes;
- one supported feature-map claim;
- one input intervention;
- why deeper does not automatically mean better.

Submit before lunch.

17 | Curve Detective: Evidence Before Labels

For each hidden run, cite two observations:

- training trend;
- validation trend;
- gap direction;
- spikes or non-finite values;
- evaluation cadence and axes.

Then propose: **underfit** | **overfit** | **unstable optimization** | **another pipeline/mode explanation**

18 | A Remedy Is a Mechanism Hypothesis

Intervention	Expected evidence
more representative data	validation improves under same model
augmentation	plausible invariance, train-only transforms
weight decay / dropout	constrained fit with useful validation retained
smaller model	lower capacity and resource use; absolute validation matters
early stopping	best checkpoint beats final checkpoint

One major intervention. One rejection criterion.

19 | LAB-D3-03 Launch

Make It Overfit, Then Rescue It | 55 min

1. Draw the expected failure.
2. Run high capacity on tiny data.
3. Mark overfit onset.
4. Choose one rescue.
5. Predict both curves.
6. Compare on aligned axes.
7. Accept or reject the mechanism.

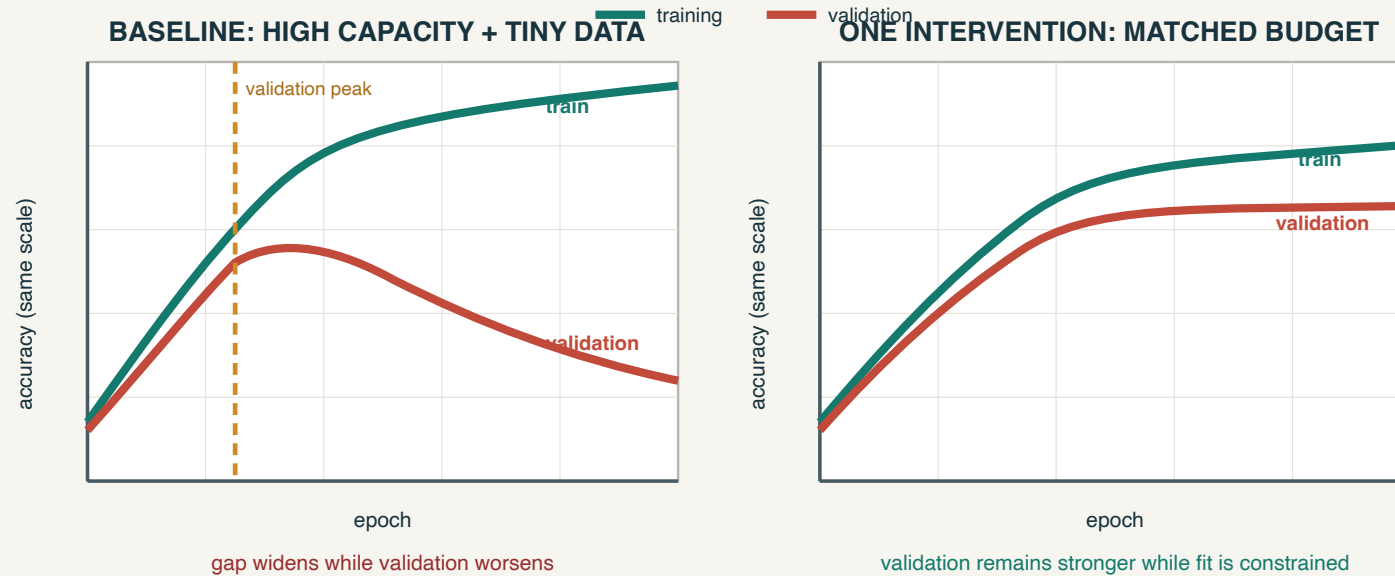
`courseware/day-3/labs/LAB-D3-03-overfit-and-rescue.ipynb`

20 | Baseline and Rescue Must Share Axes

ILLUSTRATIVE CURVES - NOT VALIDATED RUN OUTPUT

A rescue must change the evidence in the predicted direction

Same split, seed, axes, evaluation cadence, and major epoch budget.



Decision rule: a smaller gap is not a win if both train and validation collapse. Inspect absolute behavior.

A justified intervention may miss its criterion. Reject it when it does, as the validated dropout run was rejected.

21 | LAB-D3-03 Debrief: Did the Mechanism Appear?

Complete the chain:

baseline evidence -> intervention -> predicted change -> observed change -> interpretation -> next check

Ask:

- Did validation improve or did training collapse?
- Which alternative survives?
- What result would make you reverse the diagnosis?

22 | Transfer Reuses a Representation, Not Answers

Strategy	Trainable portion	Conditional value
scratch	all parameters	enough target data or strong domain mismatch
frozen features	new head	limited data/time, useful source representation
fine-tune	selected/all pretrained parameters	stable baseline plus evidence adaptation is needed

CORE: frozen features

OPTIONAL: partial unfreezing

Full fine-tuning is outside the core path.

23 | Current Torchvision Weight Contract

```
from torchvision.models import mobilenet_v3_small, MobileNet_V3_Small_Weights

weights = MobileNet_V3_Small_Weights.DEFAULT
model = mobilenet_v3_small(weights=weights)
preprocess = weights.transforms()
```

`DEFAULT` currently aliases `IMAGENET1K_V1` ; aliases may change.

Record the resolved enum. Do not use deprecated `pretrained=True` .

24 | LAB-D3-04 Launch: Choose the Source Mode

Transfer-Learning Race | 55 min

Canonical branch	Recovery branch
CIFAR-10 scratch vs frozen MobileNet V3 Small	packaged Fashion-MNIST images vs fresh head on surrogate features
<code>MobileNet_V3_Small_Weights.DEFAULT</code>	<code>transfer_surrogate_fashion_subset.npz</code>
<code>weights.transforms()</code>	<code>transfer_surrogate_embeddings.npz</code> , label-blind, 256-wide
pretrained/frozen vs random/frozen control	freeze reasoning and scoreboard practice only

**RECOVERY: NOT CIFAR-10 | NOT MOBILENET OUTPUT | NOT CANONICAL METRICS/RUNTIME | NOT PROOF
TRANSFER WINS**

`courseware/day-3/labs/LAB-D3-04-transfer-learning-race.ipynb`

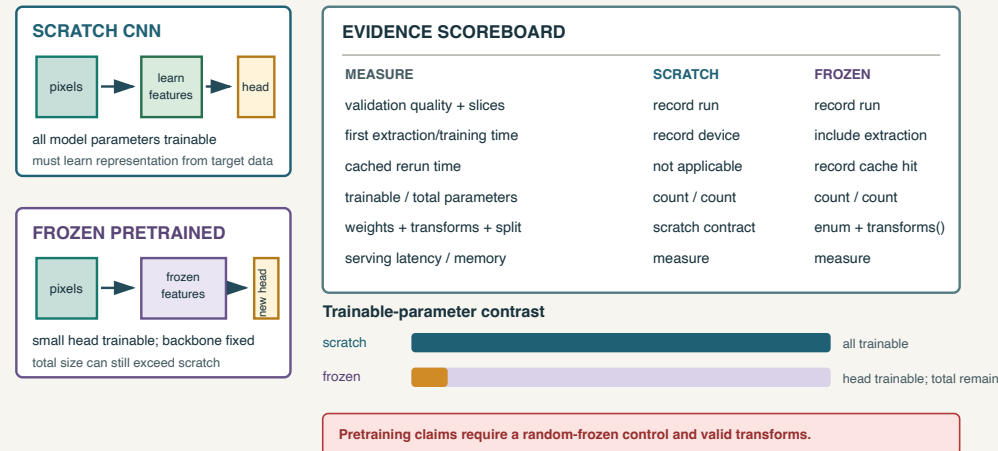
25 | Score Within the Declared Source Mode

Canonical: CIFAR-10 + explicit MobileNet weights + weight-bundled transforms.

Recovery: packaged Fashion-MNIST images + label-blind 256-wide surrogate features.

Transfer-learning race: the winner is conditional

Same split + declared budget. Record evidence before naming a winner.



**RECOVERY: NOT CIFAR-10 | NOT MOBILENET OUTPUT | NOT CANONICAL METRICS/RUNTIME | NOT PROOF
TRANSFER WINS**

26 | LAB-D3-04 Debrief: Bound the Claim

Canonical CIFAR/MobileNet: What did pretraining, random/frozen, and `weights.transforms()` establish?

Recovery surrogate: What did fresh-head training, freeze reasoning, and the scoreboard establish?

NOT CIFAR-10 | NOT MOBILENET OUTPUT | NOT CANONICAL METRICS/RUNTIME | NOT PROOF TRANSFER WINS

For either branch: **What evidence would reverse the choice? What serving constraint could change it?**

27 | ACT-D3-04: Predict Token Relevance

The dog chased the ball because it was rolling.

Focus token: **it**

Before the map:

- rank three relevant tokens/phrases;
- draw relationship arrows;
- state Q/K/V roles without equations;
- predict what missing token order would cost.

28 | Q, K, V: Three Roles in Context Mixing

- **Query:** what the current representation seeks.
- **Key:** how an available representation can be matched.
- **Value:** content carried into the weighted mixture.
- **Position:** information that helps distinguish order.

Self-attention: context is drawn from representations in the same sequence.

MOVE: attention equations and implementation

29 | Attention Weights Are Not a Complete Causal Explanation

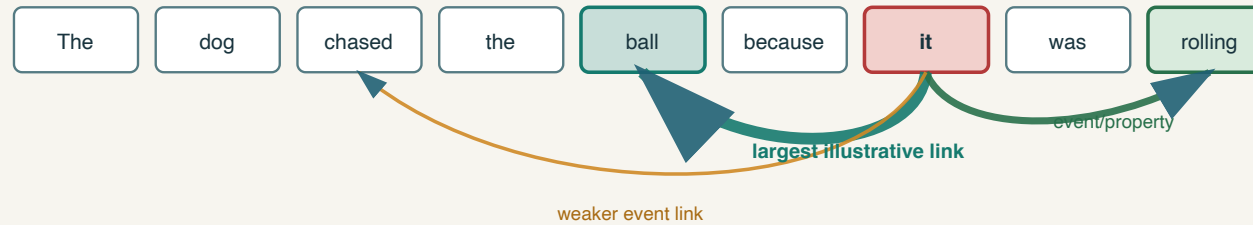
POST-PREDICTION REVEAL

Attention is one relevance path in a larger computation

ILLUSTRATIVE - NOT MODEL OUTPUT

The dog chased the ball because it was rolling.

Focus token: it

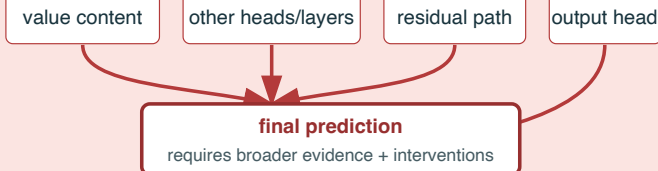


WHAT THE MAP CAN SHOW

One attention operation assigned different mixture weights to available representations.

Q: what the focus seeks
K: how context can be matched
V: content carried into the mixture

WHY THIS IS NOT A COMPLETE CAUSAL EXPLANATION



Large attention weight != causal responsibility.

30 | Transformer and Foundation-Model Connection

token + position -> self-attention -> residual/normalization -> feed-forward -> residual/normalization

Pretraining learns broadly reusable representations. Adaptation uses them for downstream contexts.

The same fundamentals remain:

inputs | parameters | activations | loss | gradients | optimization | evaluation

Boundary: conceptual connection, not foundation-model implementation mastery.

31 | Scale Turns Compute into a Bottleneck Hunt

Potential bottlenecks:

read/transform -> transfer -> compute -> communication -> checkpoint -> serve

Scale adds:

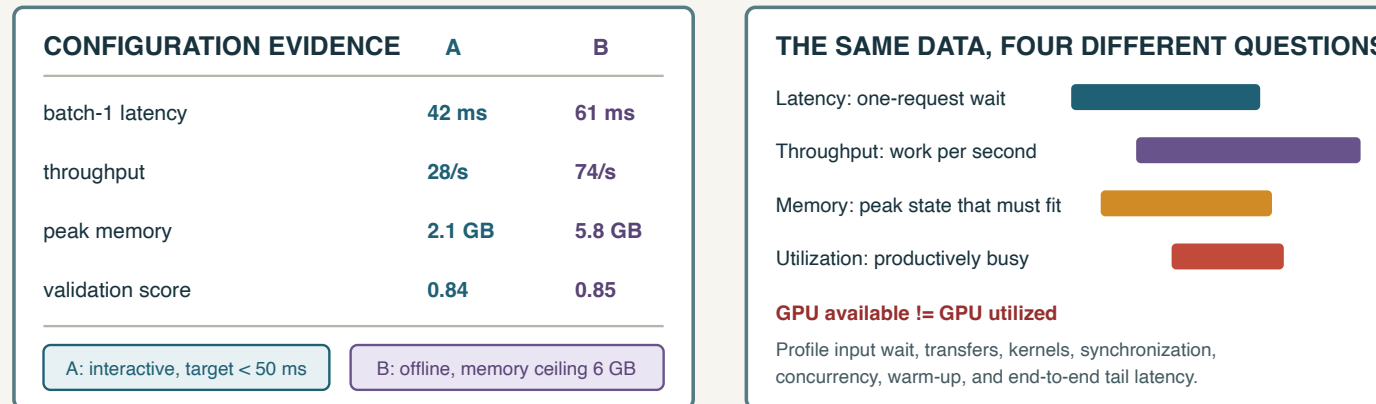
- memory pressure;
- coordination and communication;
- long-run failure/recovery;
- data-pipeline stalls;
- latency and throughput objectives.

MOVE: distributed algorithms and mixed-precision implementation

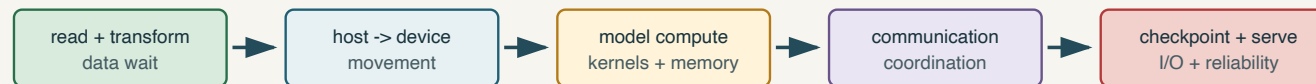
32 | Latency, Throughput, Memory, Utilization

Scale creates several scoreboards, not one speed number

Hypothetical teaching scenario. Choose from the stated constraint; remeasure on the real workload.



A bottleneck can move as workload and scale change



Prompt: Which missing measurement could reverse A or B? Tail latency, cost, slices, reliability, or concurrency?

33 | The Engineering Decision Is Conditional

Configuration A can win **latency**.

Configuration B can win **throughput**.

Neither wins every workload.

Before claiming a GPU or larger system helped, record:

- workload and device;
- warm-up and repeats;
- latency distribution and throughput;
- peak memory;
- input/transfer/compute timeline;
- quality and consequential slices.

34 | Exit Checks: Five-Minute Minimums

CHECK-D3-02 | 5 min

Name both leaks, give the repaired order, and explain why the lower honest score is better evidence.

CHECK-D3-03 | 5 min

Diagnose from two observations; choose one rescue with mechanism, confirming evidence, and rejection evidence.

Submit each live response to Day 3 Checks before moving on.

35 | CHECK-D3-04: Five-Minute Live Gate

Live: choose a strategy using data, domain, time, and trainable parameters; name the random-frozen control; then correct:

1. "The largest attention weight caused the prediction."
2. "Moving to a GPU guarantees utilization and lower latency."

Five-minute consolidation due in Day 3 Checks before Day 4 at 09:00: add switch/serving limits, a second omitted attention route, and:

claim -> evidence -> limitation -> highest-information next experiment

36 | The Fundamentals Do Not Disappear at Scale

1. Depth can reuse features; deeper is not automatically better.
2. Valid pipelines make scores interpretable.
3. CNNs expose locality, sharing, and shape.
4. Generalization needs held-out evidence and controlled rescue.
5. Transfer is conditional.
6. Attention weights are not complete causal explanations.
7. Scale adds memory, throughput, latency, communication, and reliability.

Day 4: What should we improve next, and what evidence justifies it?